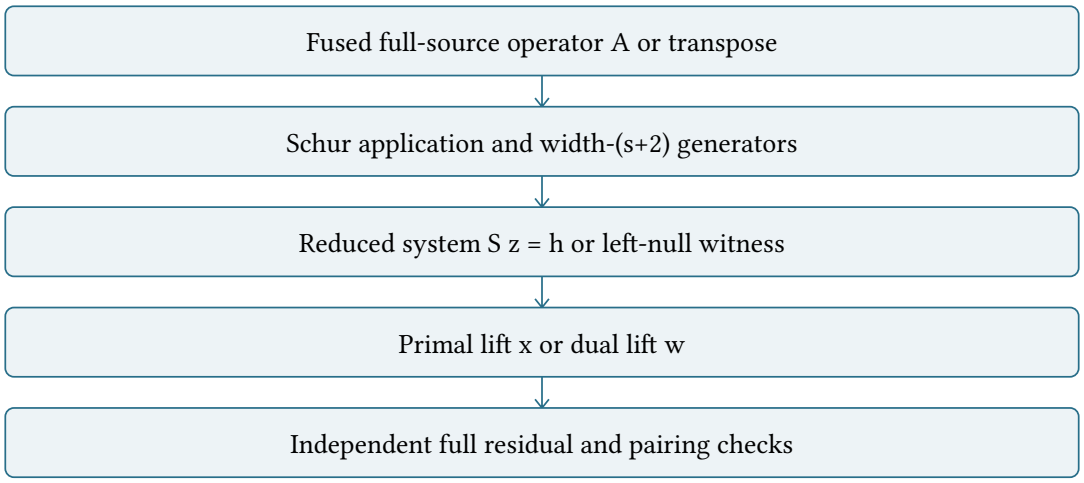
# Fused Schur adapter and certificate transport

2026-11-01 · Derived integration; syntax-checked source; zero integration trials

The adapter inherits the existing Schur construction and replaces only its full-source multiplication with the fused operator. Leading-block solves, wrap guards, correction signs and obstruction propagation remain inherited. The historical capped Schur source is unchanged. The separate full fused kernel validation is still running; integration execution has not begun.




**Exact identities.** Let A have blocks E,B,C,D, with E invertible and $S=D-C\,E^{-1}\,B$. For $b=[b1;b2]$, use $h=b2-C\,E^{-1}\,b1$. The adapters give $x=[E^{-1}(b1-Bz);z]$ and $w=[-E^{-t}\,C^{t}v;v]$. Independent reference checks verify $Ax-b=[0;Sz-h]$, $w^{t}A=[0;v^{t}S]$ and $w^{t}b=v^{t}h$, including arbitrary residuals. Thus $v^{t}S=0$ and $v^{t}h=1$ transport a normalized exclusion witness. Feature-annihilation uses the actual feature matrix in a separate check.

**Derived generator cost.** For source dimension $N=n+m$ and width $s$, the inherited generator schedule uses $q=2s+6$ full-source calls: $s+3$ in each direction. There are $s+2$ leading solves in each direction. With the fused kernel, source plus correction field products/additions/inversions are $m$, $q+m(2s+5)$, $q$. Source coefficient reductions are $qN$; native products $q(4s+8)$; packing bytes $q(3s+1)Nd$. These are derived counts, not measurements. Reference leading-oracle work remains separate: $2(s+2)n^{2}$ dense terms and $2(s+2)n$ coefficient reductions. Data-dependent native bit volumes must be summed from actual events. This adapter supplies no new leading solver.

**Full validation design.** Preserve all321 original contexts in both arms: 642 rows, including the96 full-invertible/leading-singular controls. Add zero/ones/seeded primal and dual residual checks for every valid leading block. For every singular S, construct a public normalized test from a nonzero left-null vector: choose its first nonzero coordinate j and set $h_j=v_j^{-1}$. Keep modified leading coordinates, RHS, pairing and count controls. Leading obstructions must propagate through all three lift adapters with their exact failure schedules; they do not determine full-matrix rank.

**Files and next step.** adapter.py, certificates.py and INTEGRATION.md. Both Python sources passed syntax checks only. After the active kernel panel reaches terminal verification, finish and freeze the full integration checker, analyzer, source/input locks and expanded seed schedule before trials. The42 completed historical Schur rows must never replace the full642-row plan.